



PAPER ID-310828

Printed Page: 1 of 1
Subject Code: RCAI021

Roll No:

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

MCA-INT
(SEM IX) THEORY EXAMINATION 2024-25
CRYPTOGRAPHY AND NETWORK SECURITY

TIME: 3 HRS

M.MARKS: 70

Note: Attempt all Sections. In case of any missing data; choose suitably.

SECTION A

1. Attempt all questions in brief.

2 x 07 = 14

Q no.	Question	CO	Level
a.	What are the primary objectives of cryptography?	1	K1
b.	Explain the role of substitution ciphers in classical encryption techniques.	1	K2
c.	Define and illustrate the concept of finite fields with an example.	2	K2
d.	How does the Chinese Remainder Theorem assist in cryptographic computations?	2	K2
e.	What are the challenges of message authentication in modern networks?	3	K1
f.	Differentiate between S MIME and PGP in electronic mail security.	4	K2
g.	Explain the role of encapsulating security payloads in IPSec.	5	K2

SECTION B

2. Attempt any three of the following:

07 x 3 = 21

a.	What are the principles of differential cryptanalysis?	1	K3
b.	Explain the process of key generation and distribution in the Advanced Encryption Standard (AES).	2	K2
c.	Describe the Elgamal Digital Signature Technique and its applications.	3	K3
d.	How does the Diffie-Hellman Key Exchange work in a secure environment? Provide a practical example.	4	K3
e.	Discuss the working principles of Secure Electronic Transactions (SET) and its advantages.	5	K2

SECTION C

3. Attempt any one part of the following:

07 x 1 = 07

a.	What are transposition ciphers? Explain with a practical example.	1	K2
b.	Discuss the concept of traffic confidentiality in modern cryptographic systems.	1	K2

4. Attempt any one part of the following:

07 x 1 = 07

a.	Explain modular arithmetic with examples relevant to cryptography.	2	K3
b.	How is primality testing conducted in cryptographic algorithms?	2	K3

5. Attempt any one part of the following:

07 x 1 = 07

a.	Discuss the significance of birthday attacks in cryptographic systems.	3	K2
b.	Explain the working of Secure Hash Algorithm (SHA) and its importance.	3	K2

6. Attempt any one part of the following:

07 x 1 = 07

a.	What are public key infrastructures (PKI), and how are they implemented?	4	K2
b.	Compare and contrast Kerberos Version 4 and Version 5.	4	K2

7. Attempt any one part of the following:

07 x 1 = 07

a.	Explain the principles and types of firewalls used in system security	5	K2
b.	Discuss the detection and mitigation techniques for intrusion threats in networks.	5	K2